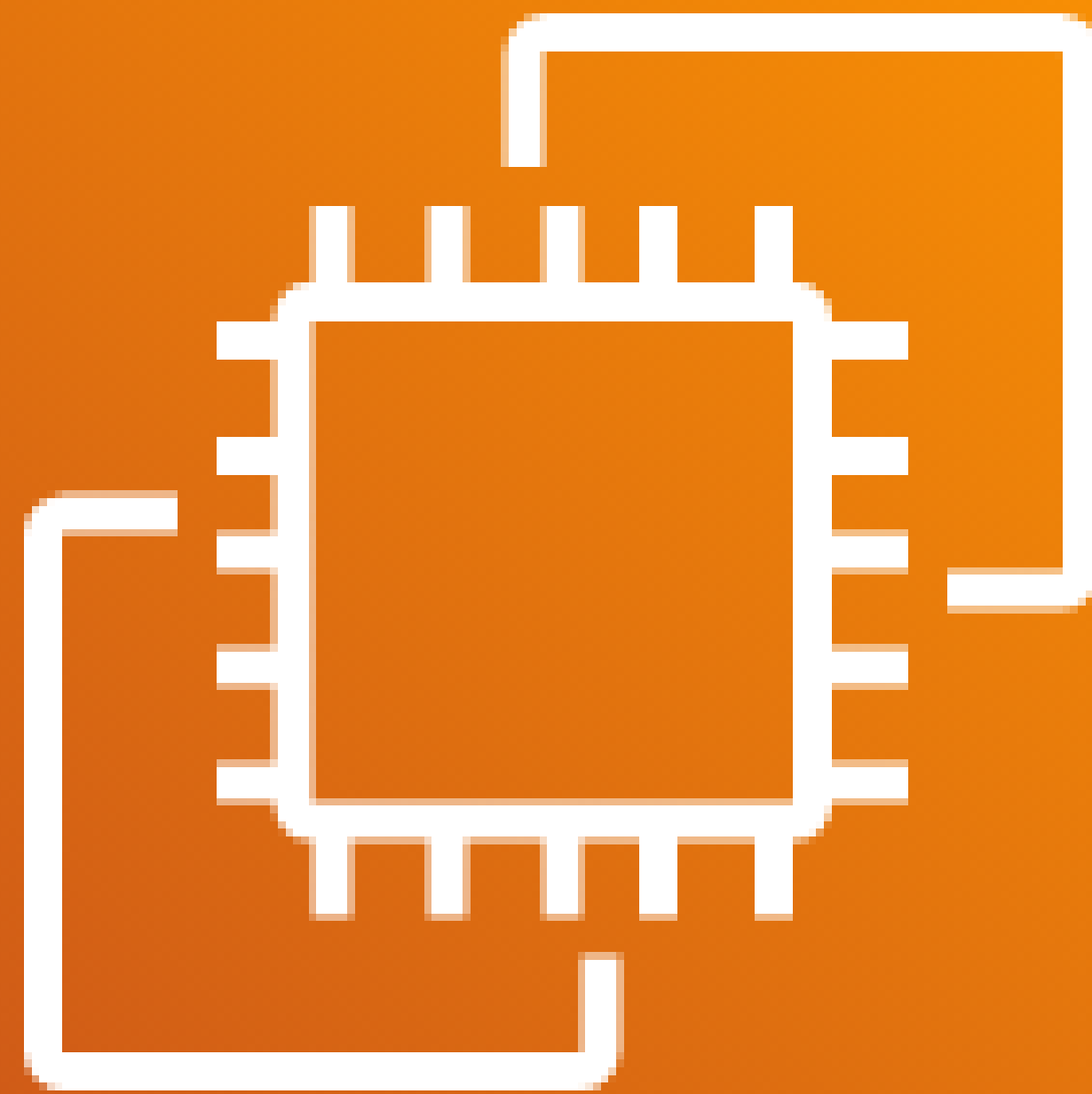


EC2



EC2 = Elastic Compute Cloud

- This is virtual machine on AWS Cloud
- It have the capability of :
 - Renting virtual machines (EC2)
 - Storing data on virtual drives (EBS)
 - Distributing load across machines (ELB)
 - Scaling the services using an auto-scaling group (ASG)

EC2 Customization Options

- OS
- CPU Count
- RAM
- Storage (Network attached - EBS, EFS / Hardware - Instance Store)
- Network card
- Firewall rules
- Bootstrap script

EC2 User Data / Bootstrap

This works only for the 1st time when we setting up instance.

Can automate boot tasks like:

- Installing updates
- Installing software
- Downloading common files from the internet

The script will run as root user

EC2 Instance Types

General Purpose

Compute Optimized

Memory Optimized

Accelerated Optimized

Storage Optimized

EC2 Instance Purchasing Options

- On demand instances
- Reserved Instances (1 year or 3 years)
- Savings plan (1 year or 3 years)
- Spot Instances
- Dedicated Hosts/Instances
- Capacity Reservations

Security Groups

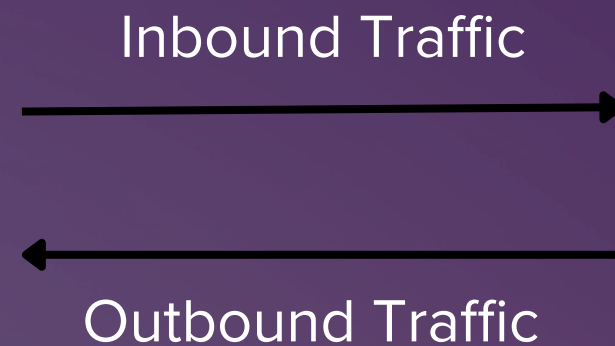
(Firewall)

Security Groups

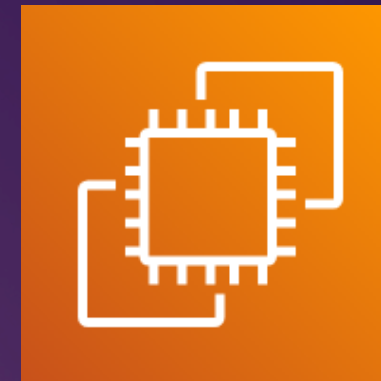
Control how traffic is allowed into or out of our EC2 Instances (like firewall)

Only contains *allow rules*

WWW



Security
Group



These decides -

- Access to ports
- Authorized IP Ranges
- Control of Inbound and Outbound Traffic

Security Groups – Important

- Can be attached to multiple instances
- Locked down to a region / VPC combination
- Does live “outside” the EC2 - if traffic is blocked the EC2 instance won’t see it
- It’s good to maintain one separate security group for SSH access
- If your application is not accessible (time out), then it’s a security group issue
- If your application gives a “connection refused” error, then it’s an application error or it’s not launched
- All inbound traffic is blocked by default
- All outbound traffic is authorised by default

Shared Responsibility Model for EC2



- Infrastructure (global network security)
- Isolation on physical hosts
- Replacing faulty hardware
- Compliance validation



- Security Groups rules
- Operating-system patches and updates
- Software and utilities installed on the EC2 instance
- IAM Roles assigned to EC2 & IAM user access management
- Data security on your instance

Practicals

**Thank
You**